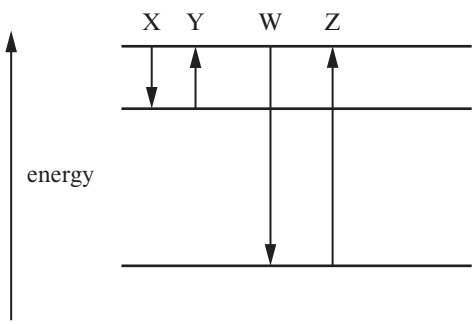
Topic E

Multiple choice questions

1 The diagram shows three energy levels of an atom.



Which transitions represent the **absorption** of a photon of the largest wavelength and which to a photon of least wavelength?

	Largest wavelength	Least wavelength
A	Z	X
B	Y	Z
C	Z	Y
D	X	W

2 A hypothetical atom has four energy levels. How many transitions in which photons are emitted are possible?

A 4
B 6
C 8
D 10

3 A photon of energy 11.2 eV is incident on a hydrogen atom in the $n = 1$ state. What is correct about the energy of the scattered photon and the final state of the hydrogen atom? The energy in the $n = 1$ state is –13.6 eV and in the $n = 2$ state the energy is –3.4 eV.

	Energy of scattered photon	Final state of hydrogen atom
A	11.2 eV	$n = 1$
B	10.2 eV	$n = 1$
C	7.8 eV	$n = 2$
D	1.0 eV	$n = 2$

4 Which of the following is **not** consistent with the Bohr model of hydrogen?

A The electron can exist in discrete energy states.

B Atomic spectra are the result of transitions between electron states.

C The difference in energy between successive energy levels is constant.

D A hydrogen atom can lose energy by emitting photons.

5 The absorption lines of visible light in the hydrogen spectrum involve transitions from $n = 2$ to higher states. Which statement is **not** correct about the effect of temperature on these lines?

A Temperature has no effect on the presence of these lines.

B There are no absorption lines at very low temperatures because most atoms are in the $n = 1$ state.

C There are absorption lines only at moderate temperatures.

D There are no absorption lines at very high temperatures because the atoms are ionized.

6 Two observations in the Rutherford scattering experiment are that most of the alpha particles went through with small deflections and very few suffered the large angle scatterings. What are the explanations for these observations in the Rutherford nuclear model?

	Reason for small deflections	Reason for large deflections
A	These alpha particles were very energetic.	The positive charge of the nucleus is much larger than that of the alpha particle.
B	The nucleus is so small that the probability that an alpha particle comes close to it is very small.	The density of the gold nucleus is much larger than the density of alpha particles.
C	The alpha particles suffer a very small number of collisions.	The alpha particles suffer a large number of collisions.
D	Most alpha particles never got close to the nucleus of the atom.	A few alpha particles got very close to the nucleus of the atom.

7 What is a typical value for the ratio $\frac{\text{size of an atom}}{\text{size of a nucleus}}$?

A 10^5

B 10^{10}

C 10^{-5}

D 10^{-10}

8 All of the following are valid conclusions from the Rutherford experiment **except** one. Which one?

A The nucleus is electrically charged.

B The nucleus is much smaller than the atom.

C The nucleus contains protons and neutrons.

D Most of the mass of the atom is in the nucleus.

9 The radius of the electron orbit in the nth state of hydrogen is n^2a_0, where a_0 is the Bohr radius. An electron in the $n = 2$ state of hydrogen has speed v. What is the speed of the electron in the $n = 3$ state of hydrogen?

A $\frac{3v}{2}$

B $\frac{2v}{3}$

C $\frac{v}{2}$

D $\frac{v}{9}$

10 The radius of the electron orbit in the nth state of hydrogen is n^2a_0, where a_0 is the Bohr radius. An electron in the $n = 1$ state of hydrogen makes f revolutions per second. How many revolutions per second does an electron in the $n = 3$ state of hydrogen make?

A $27f$
B $9f$
C $\frac{f}{9}$
D $\frac{f}{27}$

11 What are the number of neutrons and the number of electrons in the neutral atom of $^{195}_{78}\text{Pt}$?

	Number of neutrons	Number of electrons
A	117	195
B	117	78
C	195	78
D	195	117

12 An element has two isotopes. What is correct about the proton number and the nucleon number of the isotopes?

	Proton number	Nucleon number
A	Same	Same
B	Same	Different
C	Different	Same
D	Different	Different

13 Three particles are:

I protons
II neutrons
III electrons.

What particles are contained within radioactive nuclei?

A I and II
B I and III
C II and III
D I, II and III.

14 A particle has mass 1800 MeV c^{-2}. What is the approximate value of this mass expressed in u?

A 0.02 u
B 0.2 u
C 2 u
D 20 u

15 The mass difference in a nuclear reaction is 0.05 u. What is an estimate of the energy released?

A 50 MeV
B 500 MeV
C 500 J
D 5×10^{14} J